

Dynamic Electrospinning of Multimaterial Nanofiber Architectures for Programmable Metamaterials and Multifunctional Devices

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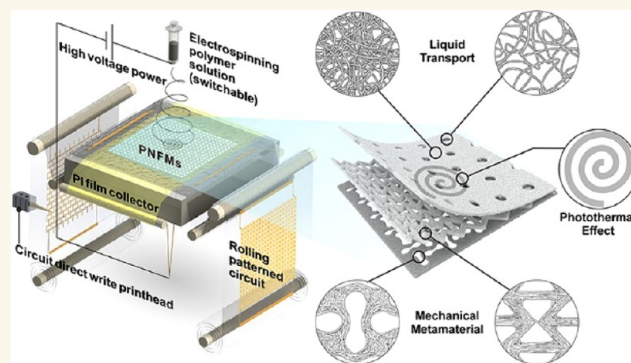
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ABSTRACT: Three-dimensional (3D) nanofiber structures are widely applicable across diverse scenarios, but their precise positioning and complex fabrication pose challenges to current 3D printing technologies. This study introduces the dynamic pattern-induced multimaterial nanofiber electrospinning technology (DPMNE Tech), an approach that integrates electrospinning with dynamic patterned circuit collectors to achieve precise nanofiber deposition. This method overcomes traditional nanofiber construction limitations by enabling the fabrication of complex, multimaterial nanofiber composite structures with microscale precision. Through the control of electric field distribution, solution properties, and pattern programming, DPMNE Tech facilitates the customization of functional mechanical metamaterials and dual-gradient directional liquid transport membranes, supporting applications in areas such as biomedical devices and liquid separation technologies. Additionally, the integration of photothermal functionalities enhances the performance of wearable patches and broadens therapeutic applications.

KEYWORDS: nanofiber metamaterials, dynamic electrospinning, patterned programmability, nanostructure, wearable patch



Three-dimensional (3D) nanofiber architectures have emerged as key structural platforms for next-generation applications in biomedicine, filtration, energy storage, and wearable electronics due to their high surface area-to-volume ratios, tunable porosity, and excellent mechanical performance.^{1–4} In biomedical contexts, nanofiber scaffolds promote cell proliferation and tissue regeneration,^{5–7} while in filtration and energy systems, they enable enhanced transport and separation capabilities due to their unique interconnected fiber networks.^{8–10} However, as advanced biomedical patches, soft robotic systems, and multifunctional textiles increasingly require materials with precisely defined spatial heterogeneity, layered structural logic, and integrated multifunctionality,^{4,11,12} the controlled fabrication of multimaterial, spatially patterned, and multilayered nanofiber structures has become a pressing bottleneck in nanoscale manufacturing.^{13–15}

Conventional electrospinning techniques, though widely adopted for their simplicity, scalability, and material versatility,^{16–20} predominantly yield randomly oriented, nonwoven fibrous mats with limited control over spatial deposition.^{21–25} While several strategies—such as rotating drums, static masks, and patterned collectors—have been introduced to induce localized fiber alignment or patterning,^{26,27} these remain

inherently static and single-material, lacking adaptability in pattern reconfiguration or material diversity during continuous processing.

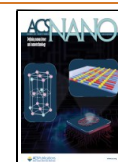
In parallel, near-field electrospinning and near-field direct writing (NFDW) have emerged as promising techniques to improve deposition resolution by minimizing the spinneret-to-substrate distance.^{28–30} These methods offer enhanced control over the fiber trajectory and pattern fidelity. However, they are fundamentally limited by their serial deposition mechanism, which renders the printing of complex infill geometries and multilayer structures highly time-consuming and computationally inefficient. Critically, NFDW exhibits poor scalability; infilling dense 2D or 3D motifs requires line-by-line tracing and may take tens of minutes to hours per square centimeter, depending on the feature density. Moreover, integrating

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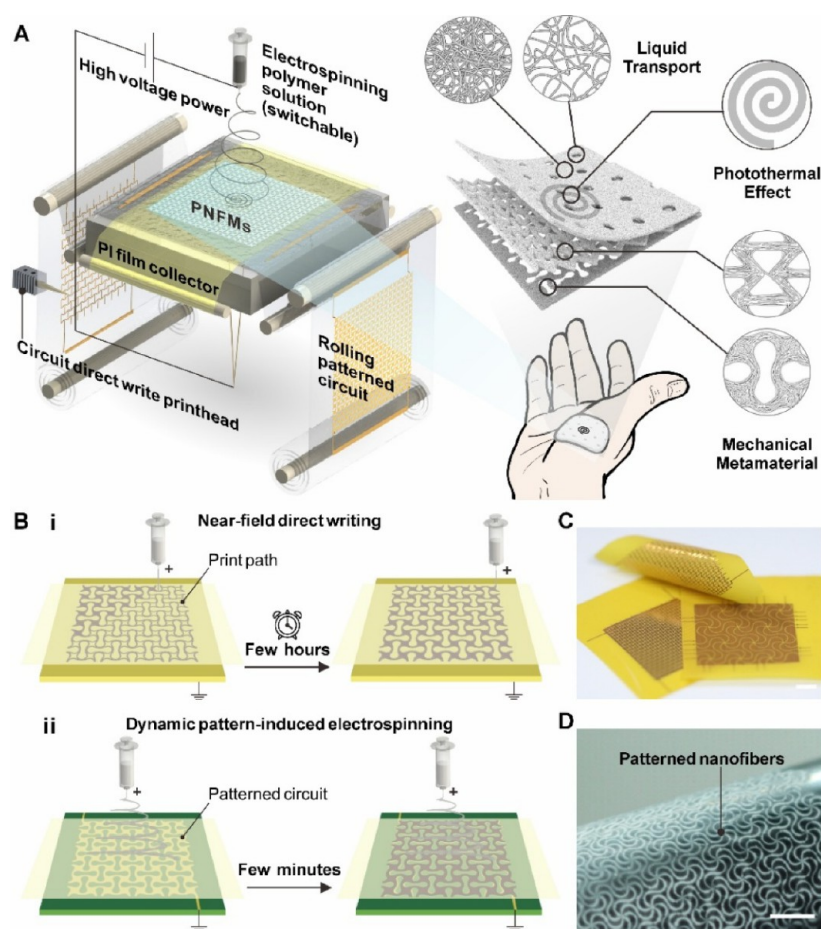


Figure 1. Demonstration of the dynamic pattern-induced multimaterial nanofiber electrospinning technology for advanced stacked patterned nanofiber membranes. (A) Overview of a schematic representation of the design and preparation process of the dynamic pattern-induced multimaterial nanofiber electrospinning technology (DPMNE Tech), as well as the construction and functionality of multipatterned composite nanofiber membranes prepared by this technology. (B) Comparison of DPMNE Tech and near-field direct writing (NFDW) technology. (C) Digital photograph of the patterned flexible printed circuit (FPC) collectors. (D) Performance demonstration of direct patterned nanofiber printing on curved surfaces using DPMNE Tech. Scale bar: 5 mm.

multiple materials in a single construct typically necessitates physical nozzle exchange or flushing protocols, introducing alignment discontinuities, increased process downtime, and a high cross-contamination risk—major obstacles for realizing fully integrated functional systems.^{29–31}

To overcome these limitations, we report a fabrication paradigm: dynamic pattern-induced multimaterial nanofiber electrospinning technology (DPMNE Tech). This platform seamlessly integrates programmable circuit-based patterned collectors, multimaterial precursor switching, and real-time electric field modulation, enabling the deposition of nanofibers into reconfigurable, multilayer, multimaterial architectures with microscale precision (Figure 1A). Unlike NFDW or static-pattern electrospinning, DPMNE Tech decouples pattern formation from nozzle mechanics by utilizing dynamic electrical addressability on the collector substrate, allowing on-the-fly pattern switching without physical translation. Furthermore, the incorporation of synchronized multimaterial delivery enables instantaneous precursor transitions, supporting the seamless integration of heterogeneous functional materials within a continuous process. Specifically, this work demonstrates the substantial potential of DPMNE Tech in advancing the capabilities of nanofiber-based materials and devices. We showcase its application across three distinct but interconnected

domains: (i) mechanical metamaterials with strain-adaptive and layer-tunable Poisson's ratios (from -0.0771 to $+0.0318$); (ii) dual-gradient Janus membranes that enable directional and gravity-defying liquid transport through combined structural and surface-wettability asymmetry; and (iii) multifunctional wearable patches that cointegrate negative-Poisson mechanical adaptability, photothermal conversion, and site-specific planar liquid-wicking pathways. These results collectively underscore the versatility and efficiency of DPMNE Tech as a unified manufacturing framework for constructing next-generation programmable micro- and nanostructures with application potential in soft robotics, biomedical therapy, intelligent interfaces, and beyond.

RESULTS AND DISCUSSION

Design and Working Mechanism of the DPMNE Tech.

DPMNE Tech combines a specific patterned circuit collector and an integrated rolling mechanism to enhance conventional electrospinning processes. Unlike conventional electrospinning setups that use a static or simple rotating drum as collectors and electrostatic direct writing methods with limited accuracy and high time consumption (Figure 1A,B), DPMNE Tech employs sophisticated micropatterned circuit collectors that can dynamically change patterns during the electrospinning process

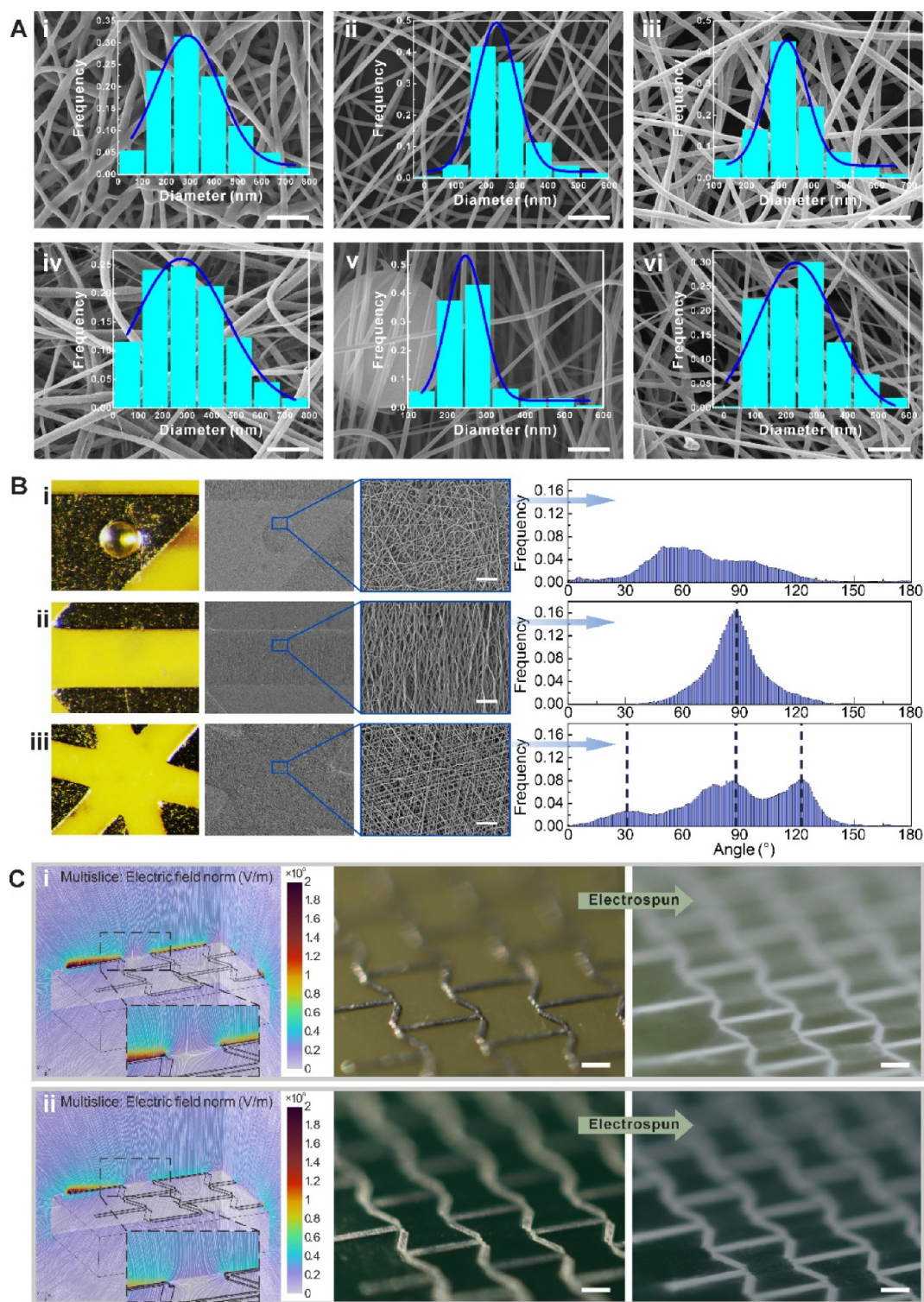


Figure 2. Precision analysis of DPMNE Tech printing. (A) SEM morphology and fiber diameter distribution of nanofiber networks from different precursor materials: (i) TPF, (ii) PVA, (iii) MWCNT@PVA, (iv) PVDF, (v) PAN, and (vi) PCL. Scale bar: 1 μm . (B) Impact of collector patterns on fiber orientation: (i) SEM image and corresponding frequency graph of fiber angle distribution for nanofibers electrospun using a circular hole collector. (ii) SEM image and corresponding frequency graph for fibers using a parallel line collector. (iii) SEM image and corresponding frequency graph for fibers using a hexagonal rotating collector. Scale bar: 5 μm . (C) Finite element analysis (FEA) and electrospinning outcomes for different collectors: (i) Re-entrant honeycomb FPC collectors and (ii) re-entrant honeycomb PCB collectors. Scale bar: 1 mm.

and are suitable for 3D stacking of nanofibers on patterned flexible printed circuit (FPC) collectors or flexible surfaces (Figure 1C,D). The initial step before sample preparation

involves using electric field simulations to predict and optimize the fiber flight path and deposition pattern (Figure 2). This is crucial for achieving the desired precision in fiber placement,

which is affected by factors such as the electric field strength, the distance between the needle and the collector, and the viscosity and conductivity of the polymer solution. Once the optimal conditions are established, dynamic pattern control comes into play. Through a controlled software interface, the pattern on the circuit collector is changed according to the desired design. This dynamically alters the circuit paths to modulate the electric field distribution on the collector's surface, guiding the deposition of the nanofibers into precise and complex geometries.

Precision is a vital metric for any printing technique because it significantly influences the performance and usability of the final materials in various settings. Achieving high precision in nanofiber 3D structures requires meticulous control throughout the electrospinning process, including the careful selection of spinning precursor materials and substrates, as well as the optimization of the electrospinning parameters and the patterned collector. This study began with a detailed analysis of fiber deposition by first examining the SEM morphology and fiber diameter distribution of nanofiber networks derived from various precursor materials, as shown in Figure 2A. This initial assessment provided a foundational understanding, setting the stage for a more targeted investigation into the deposition conditions specific to different materials. Stability and practicality are also key objectives in the development of DPMNE Tech-structures. Therefore, to balance the resilience and strength of precursor materials while ensuring spinnability, we employ a blended system of resilient polyurethane (TPU) and rigid polyacrylonitrile (PAN) materials,³² with Pluronic F127 (F127) incorporated to enhance spinning stability and improve the hydrophilicity of the final fiber/membrane.^{33,34} Figures 2A and S1 demonstrate that TPU-PAN-F127 (TPF) exhibits optimal spinning performance, featuring uniform macroscopic membrane distribution and a microfiber network free of beading, while simultaneously exhibiting synergistically enhanced toughness and strength compared to TPU and PAN. Based on this, we selected this thermoplastic TPF precursor for our case study and conducted further experiments.

To enhance the real-time switching capabilities of patterned collectors, we incorporated a polyimide (PI) film as the substrate to facilitate the layer-by-layer electrospinning of intricate 3D patterned nanofiber membranes (PNFMs), as shown in Figure S2. To quantitatively assess the quality and fidelity of the fabricated PNFMs—where fidelity is determined by precision and variability—we examined the morphological distinctions between the top and bottom surfaces of the PNFMs by microscopy after spinning and measured the nanofiber beamwidths at nine different test areas to calculate the precision and its variability. Precision is defined as the ratio of the average between the target collector's beamwidth and the measured spinning beamwidth, while variability reflects the deviation range of the spinning beamwidth. Through electrostatic field simulations paired with experimental electrospinning, we observed that 6 μm -thick PI film is most suitable for dynamic PNFM collection due to its optimal fidelity. Both simulated and experimental beamwidths of $198 \pm 4 \mu\text{m}$ were well matched (deviation of 0.6%/0.8% for the top/bottom surfaces) to the target beamwidth of 200 μm , corresponding to >99.1% precision and <2.1% variability for the FPC honeycomb collector (Figures S3 and S4). This value is significantly better than the deviation of 14.2%/16.2% for direct collection and 18.1%/14.4% for 8 μm PI film. Further investigations focused on the influence of critical electrospinning parameters, such as voltage, collection distance, and solution flow rate, on the dimensional accuracy of the

nanofiber membranes. Our research confirms that improper voltage settings can disrupt the formation of the Taylor cone that is essential for effective nanofiber production. Similarly, the optimal collection distance and flow rate are equally crucial. Therefore, careful adjustment of voltage, collection distance, and flow rate is key to achieving precise fiber deposition and high-quality nanofiber morphology.³⁵ To explore the optimal electrospinning parameters, we measured and calculated the average width and variability of the nanofiber beams at nine different test areas for each parameter setting. Balancing the voltage, collection distance, and flow rate at 16 kV, 150 mm, and 0.5 mL h⁻¹, with a 60 s electrospinning duration, uniform $200 \pm 6 \mu\text{m}$ spinning beams were obtained with a spatial variability of less than 3.4%, indicating that the template dimensions could be reproduced with high precision (>97.0%) and that the fiber network morphology was free of beading and well distributed (the detailed parameter optimization investigations are described in Figures S5–S7, and the optimal spinning settings for examined precursor solutions are provided in Table S1). Furthermore, to assess the scalability of pattern resolution, we extended the size sweep using re-entrant honeycomb collectors with target beamwidths of 27.2 μm , 71.6 μm , and 1120 μm (average), keeping the operating conditions fixed. As shown in Figure S8, fidelity was reported using two metrics from $n = 9$ readings per sample: precision and variability. When upscaled to 1120 μm , the measured beamwidth was $1135.2 \pm 16 \mu\text{m}$, corresponding to a precision of >98.6% with a variability of <1.4%. When downscaled to 71.6 μm , the system maintained >91.8% precision with variability <4.5%. At the smallest tested size (27.2 μm), the electrospun beams measured $29.7 \pm 2.8 \mu\text{m}$, giving a precision of >87.2% with a variability of <9.3%. Overall, fidelity remains high from the millimeter regime down to tens of microns—with a gradual, quantifiable roll-off only at the smallest features. In addition, the reproducibility assessment results (see Figure S9 and Supplementary Text for details) align closely with the specification of a >97.0% precision and <3.4% variability rate at the 200 μm size, demonstrating DPMNE Tech's excellent reproducibility and potential for scale-up.

To clarify how collector geometry influences fiber alignment and the observed bridging that forms between neighboring electrodes, we first explored three typical patterns—circular holes, parallel lines, and rotated hexagons. Scanning electron microscopy (SEM) and orientation analysis of fibers around circular holes exhibited a broad angular distribution without a dominant peak (Figure 2B(i)), those on parallel-line electrodes primarily aligned at 90° (Figure 2B(ii)), and those on rotated-hexagon electrodes displayed three preferred orientations at around 30°, 90°, and 120° (Figure 2B(iii)). This comparison indicates that electrode layout not only dictates the primary alignment directions but also directly results in interelectrode bridging during electrospinning. Building on these observations, we further examined more complex re-entrant honeycomb collectors through both experiments and electrostatic field simulations to understand how electrode spacing and edge effects impact bridging phenomena (Figure 2C(i)). The results revealed that the fringe fields of FPC collectors still allow fibers to span across adjacent electrodes, compromising pattern fidelity. To address this issue, we developed a specially designed printed circuit board (PCB) collector (Figure S10) that incorporates a raised solder-mask structure to locally reduce edge field intensity (detailed design process in Supplementary Text). The low-permittivity mask effectively blocks and smooths the sharp edge of the underlying metal trace, reducing the local

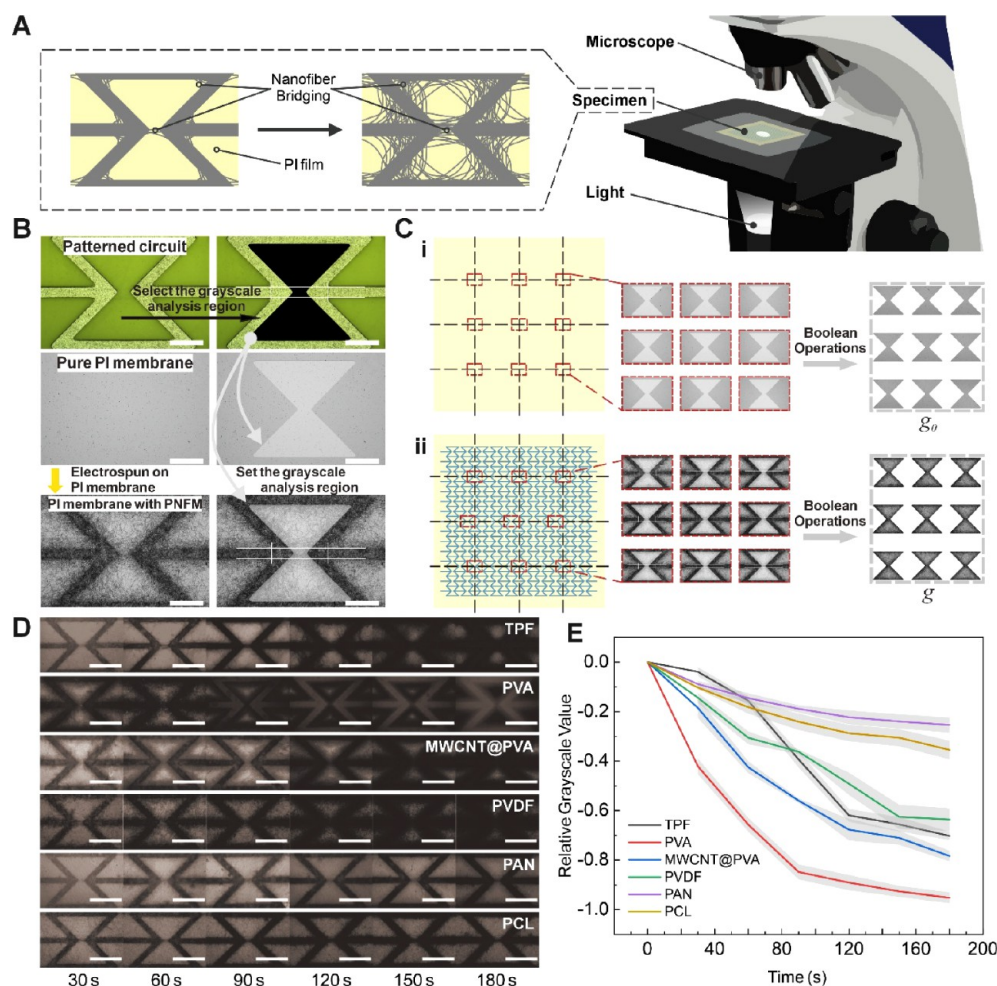


Figure 3. Evaluation of nanofiber deposition precision using grayscale analysis. (A) Schematic of the sample testing environment and nanofiber deposition. (B) Process of selecting the grayscale analysis regions. Scale bar: 500 μm . (C) Boolean operations are applied to extract the analysis regions and determine the initial grayscale value and the postspinning grayscale value. (D) Microscopic images illustrate the deposition of fibers over time using different precursor materials. Scale bar: 1 mm. (E) Time-gray level ratio curves for various precursor materials, with the light gray areas under the lines representing the error bands, demonstrating the correlation between deposition duration and gray level ratio.

normal field component and moderating the edge gradient. The associated tangential field acting on the charged jet is therefore reduced, so fibers follow the intended field lines into the patterned wells instead of bridging adjacent electrodes. Finite-element maps focused on the central gap (Figure 2C) show that the peak field at the electrode perimeter decreases from $1.3 \times 10^6 \text{ V m}^{-1}$ to $0.6 \times 10^6 \text{ V m}^{-1}$ with the solder mask between the electrodes. The effectiveness of this design is demonstrated in Figure 2C(ii), confirming that this design effectively suppresses bridging and sharpens deposition boundaries, providing enhanced control for high-precision PNFM fabrication.

To investigate how different precursor materials influence nanofiber deposition uniformity and bridging behavior, we compared their distribution profiles on a re-entrant honeycomb FPC collector overlaid with a 6 μm PI film, which allowed clean membrane release and subsequent analysis (Figure 3A). To quantify deposition uniformity, we performed a grayscale analysis on the images of deposited nanofibers (the degree of light transmission in nonpatterned regions was counted as grayscale values upon microscopic capture). We defined the relative grayscale value G (eq 1) to characterize deposition coverage:

$$G = \frac{g - g_0}{g_0} \quad (1)$$

where g_0 represents the initial grayscale value of the funnel-shaped hole area recorded before spinning (no fibers present) and g represents the grayscale value recorded after the designated spinning time. Illustrations of the grayscale region selection and Boolean operations used for determining g_0 and g are shown in Figure 3B,C. A relative grayscale value G closer to -1 indicates a higher degree of coverage in nonpatterned areas.

To better compare how different precursor materials influence fiber deposition behavior, we reflect on the deposition precision performance and variations of various nanofibers during the spinning process through the changes in G values. We systematically explored how deposition precision changed with spinning time for different materials by periodically controlling the electrospinning duration from 30 to 180 s at 30 s intervals. Each layer with different deposition times was then examined under a 50 $\times$ optical microscope, as shown in Figure 3D. The G value changes of each precursor material, along with their corresponding error bands, are detailed in Figure 3E, with narrower error bands indicating a more uniform and consistent deposition.

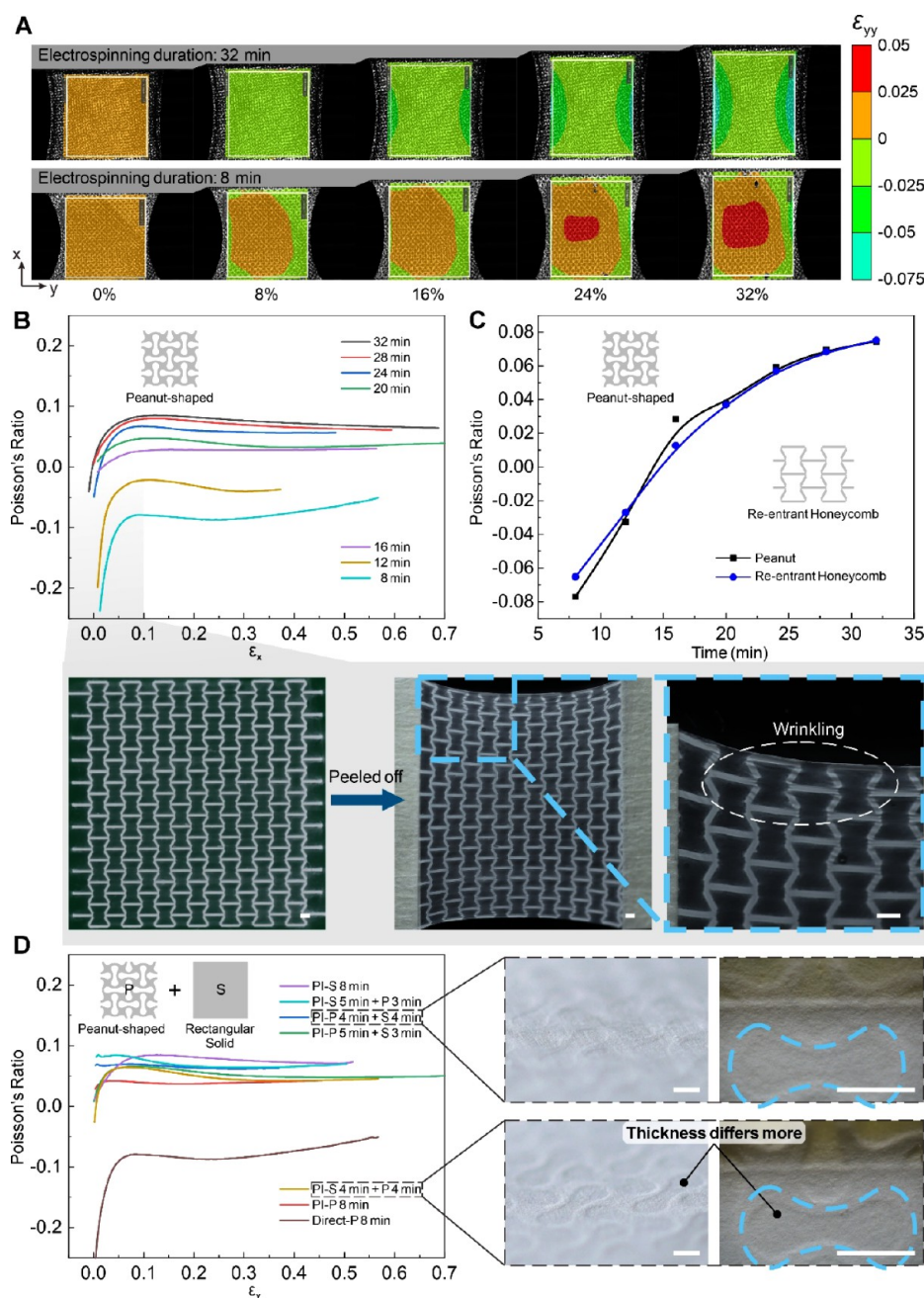


Figure 4. Mechanical properties of DPMNE Tech-produced mechanical metamaterials. (A) The variation of transverse strain (ϵ_{yy}) in the y -direction with increasing longitudinal strain (ϵ_x) in the x -direction for peanut-shaped nanofiber-based mechanical metamaterials for 32 and 8 min utilizing the DIC test system. (B) The relationship between Poisson's ratio and longitudinal strain for peanut-shaped mechanical metamaterials with various electrospinning durations. Below: PNFMs samples before (left) and after peeling off (middle and right), showing edge wrinkling postpeeling. Scale bar: 1 mm. (C) Comparison of the stabilized Poisson's ratio values for peanut-shaped and re-entrant honeycomb-shaped nanofiber-based mechanical metamaterials over various electrospinning durations. (D) The relationship between Poisson's ratio and longitudinal strain for nanofiber-based mechanical metamaterials produced with different patterned receiving modules and their respective electrospinning durations. Microscopic images show the surface texture differences for PNFMs with different sequences of patterned receiving collectors. The nanofiber-based mechanical metamaterials with a later stage of negative Poisson's ratio pattern electrospinning exhibit greater surface thickness variation. P represents peanut-shaped, S represents rectangular solid. Scale bar: 1 mm.

Our results showed that all precursor materials exhibited a decrease in deposition precision with increasing spinning time (reflected in declining G values), but the rate and magnitude are material-dependent (Figure 3D,E). This primarily stems from differences in solution conductivity, viscosity/elasticity, dielectric response, and solvent volatility, which together govern jet stability, whipping amplitude, and charge dissipation during

flight and deposition.³⁶ For instance, TPF exhibited the most controlled fiber deposition during spinning, with G values only decreasing from 0 to -0.04 at 60 s and to -0.70 at 180 s, while maintaining more uniform fiber coverage. Poly(vinyl alcohol) (PVA) had the fastest deposition rate, but its deposition precision decreased most rapidly, with the G values dropping from 0 to -0.42 at 30 s and reaching -0.95 at 180 s. Multiwalled

carbon nanotube doped poly(vinyl alcohol) (MWCNT@PVA) and polyvinylidene difluoride (PVDF) followed similar intermediate trends, with G values of -0.19 and -0.14 at 30 s and -0.78 and -0.64 at 180 s, respectively, indicating more rapid infill than TPF but also increased variability due to electrostatic interactions with the applied electric field. In contrast, PAN and polycaprolactone (PCL) were deposited more slowly and steadily, with G values of only -0.25 and -0.36 at 180 s, showing minimal deviation throughout, suggesting more consistent but less complete coverage. These results validate TPF's balance of precision and deposition rate, highlight PVA's speed advantage despite early bridging, and establish PAN and PCL as materials offering high pattern fidelity, forming a reliable basis for subsequent development of multimaterial, multipatterned, and multifunctional nanofiber 3D constructs.

To accommodate continuous production and larger fabrication, the system integrates a rolling mechanism. The patterned circuit collectors are part of a roll that can move synchronously with the electrospinning process, ensuring the consistent deposition of nanofibers over extended areas and facilitating the processing and stacking of the electrospun membranes (Figure S11). Notably, this dynamic patterned electrospinning process of DPMNE is not subject to the shortened spinneret distances and equipment motion speeds that limit production and deposition speeds, as is the case with the near-field direct writing (NFDW) method. By simulating the dense stacking of the same peanut-shaped patterned nanofibrous membranes by the NFDW method with typical parameters, we found that the NFDW method takes about 8.7–20 h to form the same patterned area and thickness as DPMNE Tech in 8 min, representing a speed difference of ~ 500 – 1100 times (detailed descriptions and analysis are shown in Figure S12 and the Supplementary Text). The comparison of mechanical properties between structures prepared by DPMNE Tech and those produced by conventional methods (NFDW and molding) demonstrates that, in addition to advantages in precision and efficiency, the nanofiber structures fabricated via DPMNE exhibit outstanding mechanical properties (Figure S13), demonstrating the highest tensile strength (1.63 MPa, more than twice that of NFDW- and molding-derived structures) and considerable toughness (relatively large failure strain). Moreover, the capability to dynamically switch patterns allows for the sequential and alternating deposition of different materials, enabling the creation of complex multimaterial structures. The technology's multimaterial capability is facilitated by multiple switchable syringe pumps that can deliver different polymer solutions to a single needle system. This setup, combined with dynamic pattern control, allows for the layering or mixing of materials within a single membrane structure, expanding the functional capabilities of the resulting nanofiber membranes.

For materials with varied functional requirements, the DPMNE platform offers programmable switching of precursor materials, patterned collectors, and deposition sequences (Figure 1A), enabling: (i) mechanical metamaterials whose Poisson's ratio can be tuned layer-by-layer; (ii) Janus membranes with dual structural-wettability gradients for directional liquid transport in biomedical and microfluidic devices; and (iii) multipattern, multilayer films incorporating precursor-specific electrospun fibers that exhibit selective responsiveness to different light wavelengths, including QR-code patterns for anticounterfeiting (Figure S14 and Movie S1). Together, these capabilities establish DPMNE Tech as a

versatile platform for next-generation functional nanofiber architectures.

Mechanical Metamaterials through DPMNE Tech. In order to demonstrate the capabilities of DPMNE Tech in the development of mechanical metamaterials, we utilized strategically designed patterned PCB collectors, including “peanut-shaped”³⁷ and “re-entrant honeycomb” designs. These collectors optimized nanofiber deposition and enhanced the mechanical properties, enabling precise tailoring of features such as Poisson's ratio at the microscale. First, we explored the influence of nanofiber layer thickness on Poisson's ratio of metamaterials by leveraging precise control over the electrospinning process duration. To rigorously evaluate Poisson's ratio of these metamaterials, we integrate digital image correlation (DIC) techniques with a tensile machine for precise real-time strain monitoring (Figure S15). The collected data meticulously track variations in Poisson's ratio in response to longitudinal strain (Figure 4A and Movie S2). Statistical analysis reveals that samples with shorter electrospinning durations (8–12 min) or thinner nanofiber depositions exhibit fluctuations in Poisson's ratio at lower strain levels (less than 10%). Beyond this threshold, Poisson's ratio stabilizes and clearly manifests as a distinct negative value ($-0.0771 \sim -0.0326$ for “peanut-shaped”, $-0.0652 \sim -0.0270$ for “re-entrant honeycomb”). Conversely, samples subjected to longer electrospinning times (more than 16 min) or thicker nanofiber depositions demonstrate a positive Poisson's ratio beyond 10% strain level (Figures 4B and S16).

Electrostatic field simulations further illustrate this phenomenon (Figure S17). The electrostatic field distribution and local magnification for the peanut-shaped and the re-entrant honeycomb-patterned collectors reveal that as the thickness of the nanofiber layer increases at initial, 8 min, and 16 min of electrospinning, the field strength above the metal electrodes gradually weakens, resulting in more disordered nanofiber deposition, which consequently leads to the observed increase in Poisson's ratio. Furthermore, the significant variations in Poisson's ratio observed in both thicker and thinner metamaterial samples at low strain levels stabilize upon reaching a certain strain threshold. At low axial strains, the apparent fluctuations in Poisson's ratio mainly arise from peel-induced residual stress and the ensuing wrinkle flattening. After detachment from the collector, residual compressive stress produces edge wrinkling and warpage (white circles in Figure 4B). Under small stretching, the deformation first planarizes these wrinkles as their amplitude decreases with strain, so the measured lateral strain is governed by geometric unwrinkling rather than auxetic hinge rotation. This leads to a nonlinear, sometimes sign-changing, lateral response. Once the wrinkles vanish at a threshold strain and the fibers align, the deformation switches to unit-cell behavior: hinge rotation dominates in thinner layers (negative ν), whereas ligament stretching dominates in thicker layers (positive ν). Poisson's ratio then stabilizes, consistent with Figure 4B and the ~ 0 – 10% strain regime noted in the multilayer tests. We quantified the stabilized values of Poisson's ratio for the above metamaterials fabricated at varying thicknesses or electrospinning durations (Figure 4C). Our data demonstrate that, with identical patterns, we can manipulate the thickness of the electrospinning to control Poisson's ratio of the metamaterials and obtain patterned nanofibrous membranes with negative Poisson's ratio properties. It has been shown that negative Poisson's ratio network materials are able to better fit complex curved surfaces such as

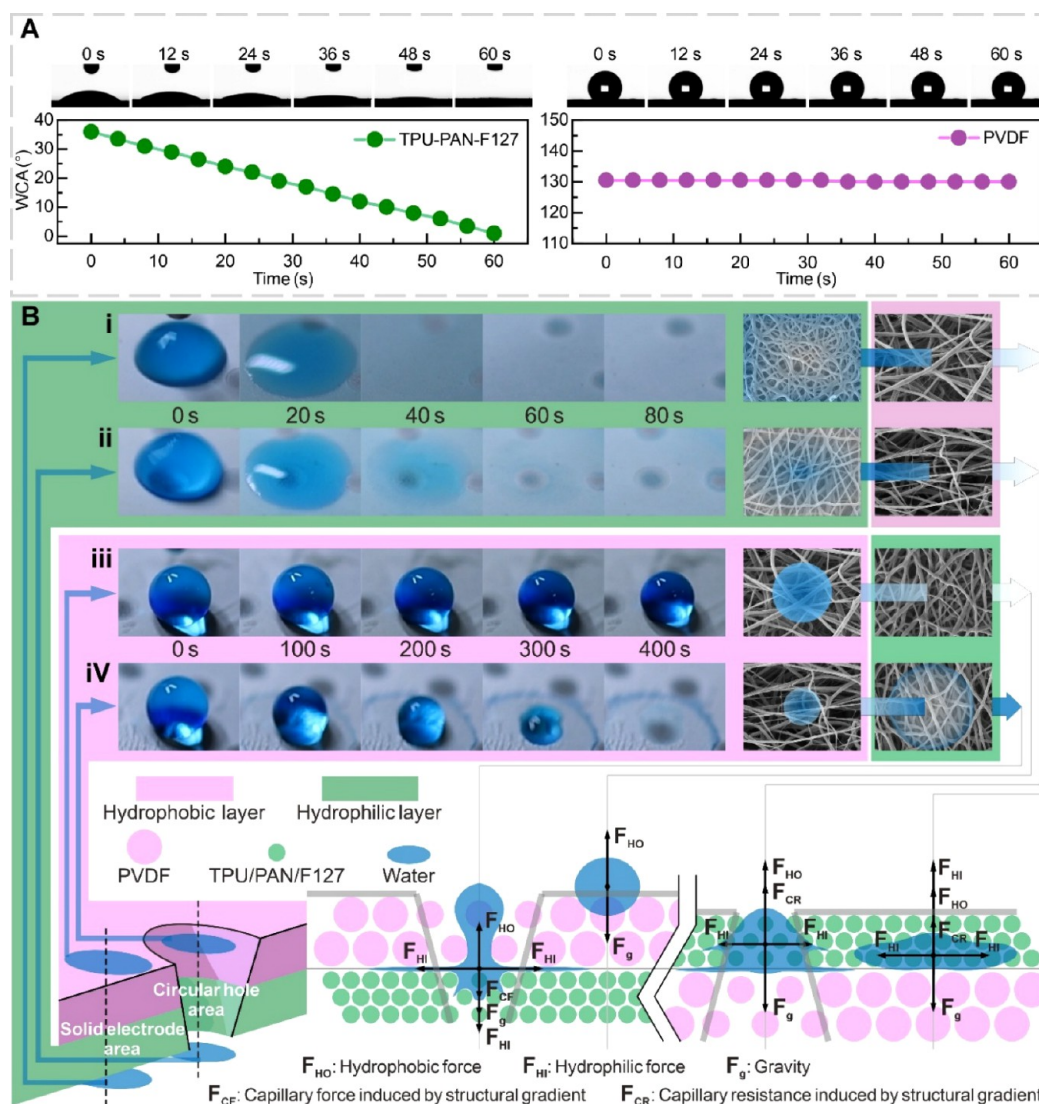


Figure 5. Testing of dual-gradient directional liquid transport membranes. (A) Time series of water contact angle (WCA) measurements for the hydrophilic TPU-PAN-F127 electrospun membrane and the hydrophobic PVDF electrospun membrane. (B) Visual documentation of dyed water droplet movement on the solid electrode areas and circular hole areas of the TPU-PAN-F127 and PVDF membranes, complemented by SEM images of these areas and a detailed diagram explaining the transport mechanisms. (i) Water droplets on the solid electrode area of the TPU-PAN-F127 side. (ii) Water droplets on the circular hole area of the TPU-PAN-F127 side. (iii) Water droplets on the solid electrode area of the PVDF side. (iv) Water droplets on the circular hole area of the PVDF side. This schematic illustrates how hydrophobic forces (F_{HO}), hydrophilic forces (F_{HI}), capillary forces from structural gradients (F_{CF}), capillary resistance (F_{CR}), and gravity (F_g) interact to drive the liquid transport process.

human skin due to their unique isotropic deformation mechanism, which reduces wrinkles or voids during dynamic strains and simultaneously delays material fatigue and delamination, resulting in enhanced performance and comfort for biointegrated devices.³⁸

Further expanding our analysis to multiple pattern printing effects on Poisson's ratio, we investigate the integration of patterned collector module transitions in TPU-PAN-F127 multilayer electrospinning. Figure S11 illustrates the alternating use of the peanut-shaped PCB collector and the rectangular solid PCB collector. In this study, we observed a phenomenon similar to that seen in single-pattern printing, where Poisson's ratio of the fiber membranes exhibits fluctuations during the initial 10% of longitudinal stretching but gradually stabilizes as lateral strain increases. This behavior is attributed to microstructural adjustments and fiber realignment occurring during the initial stretching phase. As shown in Figure 4D, our comparative

analysis of different pattern printing ratios revealed that using either the peanut-shaped PCB collector or the rectangular solid PCB collector for 100% of the spinning time resulted in the lowest and highest Poisson's ratios observed in the experiments, respectively (PI-S 8 min represented by the brown line and direct-P 8 min represented by the purple line). When both collector types were used for an equal amount of time (50% each), placing the rectangular solid structure as the second layer on top of a layer printed with the peanut-shaped pattern increased the overall Poisson's ratio of the fiber membrane (PI-P 4 min + S 4 min represented by the dark blue line). Conversely, using the peanut-shaped pattern as the second layer resulted in a lower Poisson's ratio, indicating reduced mechanical constraint from the layer below (PI-S 4 min + P 4 min represented by the yellow line). This is due to the fact that the sample using the peanut-shaped pattern as the second layer exhibits a more distinct peanut-shaped texture, which was able to effectively

modify the mechanical properties of the resultant metamaterials, as shown in the close-up digital photo in Figure 4D. These observations suggest that Poisson's ratio of the overall multilayer patterned membrane can be modulated by precise control of spinning time and module transitions.

Detailed SEM was conducted on the cross-sectional profiles of samples to assess the impact of transitions between patterned collection modules on fiber network morphology. The configurations tested included samples collected entirely using rectangular solid PCB, a combination of 50% rectangular solid PCB and 50% peanut-shaped PCB, and samples collected entirely using peanut-shaped PCB, all spun for the same total duration. As shown in Figure S18, the samples spun exclusively with the rectangular solid PCB collector displayed uniform thickness throughout, with an average measurement of 12.5 μm . This uniformity showcases the controlled deposition environment that solid electrodes provide. In contrast, samples spun exclusively with the peanut-shaped PCB collector showed periodic variations in thickness across their cross-sections, reflecting the alternating patterns of voids and solid areas characteristic of the patterned electrodes. The thickest regions of these samples measured up to 18.2 μm , while the thinnest dipped to 0.59 μm , underscoring significant variability in thickness that plays a crucial role in their unique Poisson's ratio properties. For samples combining both rectangular solid and peanut-shaped PCB collectors equally, with the solid-plane layer spun first, the cross-section also demonstrated a periodic thickness pattern. However, unlike the pure peanut-shaped samples, the thinnest regions in these mixed samples were noticeably thicker, at 4.51 μm , approaching the 18.1 μm thickness seen in the thickest parts. This variation helps explain the differences in Poisson's ratio observed between samples spun exclusively with the rectangular solid PCB collector and those involving the peanut-shaped PCB collector, revealing the mechanical interactions influenced by the layered patterning. Overall, this work demonstrates how precise control over spinning time and module transitions can finely tune the mechanical properties of metamaterials, thus achieving precise control over Poisson's ratio.

Dual-Gradient Directional Liquid Transport Membranes. Beyond its applications in the fabrication of mechanical metamaterials, DPMNE Tech has also been pivotal in developing materials for directional liquid transport. Utilizing principles such as Laplace pressure, we created custom-designed Janus liquid transport membranes featuring dual gradients. This achievement originates from the combination of materials with varying hydrophilic properties, microstructural design, and macroscopic patterning, thus enabling precise control over liquid movement across the membrane and adjustable in-plane wettability. Unlike traditional nonspecific Janus designs, DPMNE Tech has the advantage of being able to customize the Janus feature area by patterning, i.e., specifying the location and patterns where Janus functions should be performed on the integral membrane. These capabilities are invaluable in applications that demand precise liquid separation and controlled chemical reactions within microfluidic systems. For the development of Janus liquid transport membranes, this study utilized the electrospinning precursor materials TPF and PVDF, chosen for their contrasting hydrophilic properties. TPF, known for its hydrophilic properties and excellent mechanical strength, is easily removed from the collector and serves as the hydrophilic layer in the Janus membranes.³⁹ PVDF, noted for its hydrophobic characteristics, was used for the hydrophobic

layer.⁴⁰ Both materials use *N,N*-dimethylformamide (DMF) as a solvent, facilitating strong interfacial adhesion during the spinning process. Water contact angle (WCA) measurements for the hydrophilic TPF electrospun membrane and hydrophobic PVDF electrospun membrane show the change in WCA over time (Figure 5A). TPF membranes exhibit strong hydrophilicity, with the contact angle dropping from 36.0° to 0° in 60 s, while PVDF remains hydrophobic, with only a slight decrease from 130.5° to 130.0°, indicating minimal spreading and resistance to water penetration.

We adapted our DPMNE Tech to introduce structural variations that enhance the liquid transport properties. A patterned PCB collector with circular holes was crafted, and its solder mask layers were removed to lower the circular hole areas compared to solid electrode ones, purposely creating nanofiber bridging during the printing process (Figure S19). This design facilitated fiber networks when electrospinning TPF and PVDF, and Table S2 analysis showed notable differences in fiber diameters, especially around the holes, where fibers formed bridging connections. This bridging not only yielded larger, more oriented pore structures but also thinner fiber diameters compared to those in solid electrode areas, successfully constructing a structural gradient.^{41,42} Combined with the inherent wettability gradient of the two materials, this resulted in the formation of dual gradients.

Following this, we conducted water contact angle tests in different regions to elucidate how each region of the Janus membrane governs droplet behavior (Figure 5B): (i) Hydrophilic side (the TPF layer), solid region. A droplet spreads radially and evaporates completely within 24 s, as fine TPF fibers generate strong hydrophilic attraction (F_{HI}). SEM analysis revealed finer fibers and smaller pores in this layer, in contrast to the coarser fibers and larger pores in the hydrophobic PVDF layer (Table S2). (ii) Hydrophilic side (the TPF layer), hole region. The droplet expands more slowly, taking approximately 80 s to evaporate due to the larger pore size beneath, which amplifies the capillary resistance (F_{CR}). Despite the slower rate, the hydrophobic forces (F_{HO}) prevented the droplets from penetrating into the hydrophobic layer, ensuring that the droplet fully evaporates within the TPF layer without reaching the PVDF layer. (iii) Hydrophobic side (the PVDF layer), solid region. It maintains a nearly spherical profile for over 400 s, primarily losing volume through slow evaporation. The dense PVDF network and upward surface tension (F_{HO}) effectively block lateral spreading. (iv) Hydrophobic side (the PVDF layer), hole region. In this region, the droplet permeates through the porous PVDF structure within 372 s, contacts the underlying TPF layer, and spreads out to form a characteristic "coffee-ring" effect. Complete evaporation occurs from the hydrophilic layer, while hydrophobic drag (F_{HO}) limits the liquid transport rate in the upper layer, resulting in a confined wet area. Together with the structural (pore-size) gradient introduced by the patterned regions, these results demonstrate that the coupled wettability gradient and pore-size distribution establish a Laplace-pressure difference (Supplementary text, eq S1) that enables precise, region-specific liquid control. When liquid is introduced at the hydrophobic-hole region, which is aligned with the direction of gravity, capillary forces (F_{CF}) and hydrophilic attraction (F_{HI}) overcome both gravity and hydrophobic resistance (F_{HO}) to pump fluid upward into the hydrophilic TPF layer, effectively achieving antigravity liquid transport, as shown in Movie S3. Consistent with the estimate in (Supplementary text eqs S2–S5 and Table S2), the net capillary driving pressure ΔP is orders of

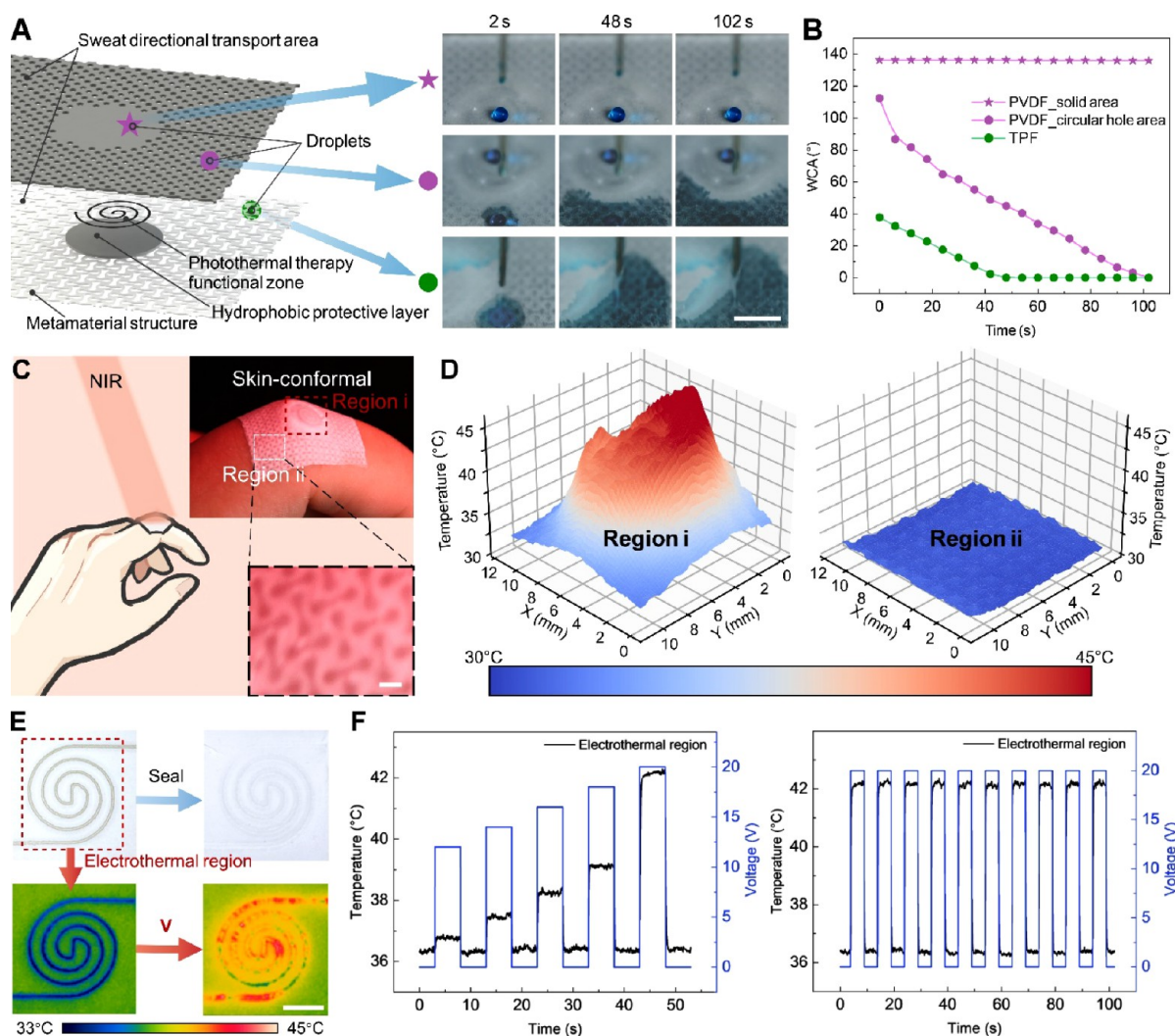


Figure 6. Performance and structure of multifunctional wearable patches. (A) Structural schematic of multifunctional wearable patches and liquid transport over time in different functional zones of the patches. Scale bar: 10 mm. (B) Water contact angle change over time for different functional zones of multifunctional wearable patches. (C) Schematic of photothermal heating and liquid management of the patches. Scale bar: 5 mm. (D) Photothermal performance in regions i and ii with temperature distribution maps. (E,F) Structure of printed flexible patterned circuits and their electrothermal properties. Scale bar: 5 mm.

magnitude larger than ρgh for millimeter-scale heads, so gravity is negligible in comparison.⁴³ Consequently, the PVDF side remains only locally wetted, while the TPF side rapidly spreads and evaporates the liquid, evidencing the directional and efficient routing of moisture across the bilayer.

Design and Performance of Multifunctional Wearable Patches. Utilizing DPMNE Tech's capabilities for multi-material, multipattern nanofiber integration, we have developed multifunctional composite PNFMs that synergistically combine photothermal functionality, negative Poisson's ratio mechanical properties, and directional liquid transport. This integration is particularly promising for applications in biomedical fields, where such multifaceted functionalities can enhance therapeutic outcomes. As shown in Figure 6A, the working mechanism of our fabricated multifunctional wearable patches is displayed, which includes the metamaterial structure for adaptive surface conforming, a photothermal therapy functional zone, a sweat directional transport area, and an external hydrophobic protective layer. Using DPMNE Tech, TPF, PVDF, MWCNT@PVA (4.5 wt % MWCNT weight ratio), and

PVDF were sequentially printed in layered patterns, including peanut-shaped structures, homogeneous disc-shaped structures, spiral structures, and an array of holes (Figures 6A and S20). In the wearable functional patches, the negative Poisson's ratio metamaterial layer provides excellent conformability to curved surfaces, creating a seamless fit on curved skin surfaces, improving contact and treatment efficiency while ensuring patient comfort.³⁸

We further demonstrated the excellent liquid management capabilities of our wearable patches. As shown in Figure 6C and Movie S4, on the solid areas of the PVDF layer, water droplets exhibited minimal spreading and evaporated slowly, maintaining a nearly spherical shape due to robust hydrophobic forces. In contrast, droplets on the circular hole areas of the PVDF layer penetrated through to the underlying hydrophilic TPF layer, where they spread and evaporated quickly due to the dual effects of wettability and structural gradients. This facilitated efficient liquid management by channeling fluids away from the skin-contacting layers. When droplets were directly placed on the outer hydrophilic TPF layer, they spread rapidly and evaporated,

with the underlying hydrophobic PVDF layer acting as a barrier to prevent any leakage into the skin-contacting inner layer. This feature is critical for maintaining skin dryness and comfort under the patch. Contact angle measurements corroborate this tailored response of the patch to liquids (Figure 6B): on the hydrophilic TPF layer, the WCA drops from 37.8° to 0° within 48 s, indicating rapid spreading. On the PVDF side, the water droplet behavior depends on the geometry—droplets on the circular-hole regions decrease from 112.4° to 0° over 102 s, whereas those on the solid PVDF remain around 136° throughout the test—demonstrating precise spatial control of liquid transport within the composite membrane. As a result, the hydrophobic PVDF layers designed with an array hole pattern synergize with the hydrophilic TPF layers to create a dual-gradient directional liquid transport structure. This design allows wound exudate or sweat to drain from the skin surface more efficiently, enhancing the comfort of the wearer. Additionally, a disc-shaped waterproof PVDF layer was devised between the photothermal region and the hydrophilic TPF layers, strategically preventing fluids on the TPF layer from penetrating the photothermal therapy region to ensure that the therapeutic function of the patch is not impaired by liquid.

For the photothermal therapy functional zone, multiwalled carbon nanotubes (MWCNTs) were selected as the photothermal material due to their high photothermal conversion efficiency within the NIR-I window (700–950 nm), which offers deep penetration into biological tissues while minimizing thermal damage.⁴⁴ A photothermal nanofiber precursor was prepared by blending MWCNTs with PVA, followed by fabricating the MWCNT@PVA photothermal therapy functional nanofiber membrane using a nonpatterned collector (Figure 6C). The distribution of MWCNTs can be observed through transmission electron microscopy (TEM) characterization of the samples (Figure S21A). A UV–vis–NIR diffuse reflectance test showed that while pure PVA fibers primarily absorbed in the UV–visible range (Figure S21B), MWCNT@PVA nanofibers absorbed across the UV–visible–NIR spectrum, particularly within the NIR-I window (Figure S21C). This broadband response stems from the delocalized π -electron network in the graphene walls of the nanotubes. NIR photons promote mobile carriers that relax nonradiatively via electron–phonon coupling on a subpicosecond time scale, and the resulting lattice heat is rapidly transferred to the surrounding polymer. A photothermal conversion efficiency of $\eta = 49.14\%$ was obtained with a 4.5 wt % MWCNT doping (calculations are detailed in Supplementary Text). The heat generation could be controlled through both MWCNT doping concentration and infrared laser power (Figure S22). Then, we evaluated the patch's photothermal performance, focusing on specific areas exposed to near-infrared (NIR) radiation. The temperature profiles of these regions are shown in Figure 6D, where photothermal region i exhibits a noticeably higher average temperature of 38.6 °C with a peak of 46.2 °C, whereas reference region ii has an average temperature of 31.1 °C with a peak of only 31.9 °C, confirming the patch's capacity for targeted localized heating and controllable thermal distribution. Moreover, the photothermal region of the patch efficiently generates safe therapeutic heat, maintaining steady-state temperatures between 43 and 46.5 °C even during cyclic irradiation (Figure S23). It is worth mentioning that this technique of printing flexible patterned circuits on an electrospinning membrane can also directly endow the PNFs with electrothermal properties. As shown in Figure 6E, we printed a bihelical pattern of silver

nanowires onto a prepared electrospun membrane and connected it to a time relay and DC power to form a conductive circuit. This designed planar bihelical pattern allows the patterned area to be uniformly heated. As the voltage was gradually increased from 12 to 20 V, the sample temperature steadily elevated from an initial near-body temperature of ~36.3 °C to approximately 42.2 °C at 20 V, demonstrating that the patterned area can effectively generate safe therapeutic heat at body-safe voltages. Moreover, the sample consistently warms quickly and maintains a stable temperature during repeated cycling tests (Figure 6F). Overall, the advanced capabilities of DPMNE Tech have been crucial in developing these innovative multifunctional wearable patches. By strategically combining photothermal elements, mechanical adaptability, and controlled liquid transport mechanisms, we have provided an option for the design of wearable medical devices, offering customized therapeutic options tailored to the needs of modern medical practices.

CONCLUSIONS

In this work, we introduce dynamic pattern-induced multi-material nanofiber electrospinning technology (DPMNE Tech), a highly versatile and programmable platform enabling precise control of electrospinning over pattern geometries, material compositions, and 3D stacking through dynamic electric field modulation and multiprecursor switching. DPMNE Tech achieves microscale precision, material heterogeneity, and structural complexity in nanofiber construction while enabling rapid and customized deposition, overcoming long-standing challenges in conventional electrospinning, near-field direct writing, or additive manufacturing for nanofiber structures.

We constructed mechanical metamaterials with programmable Poisson's ratios ranging from -0.0771 to $+0.0318$ by modulating deposition time and collector sequence, providing tunable auxetic behavior ideal for conformal biointerfaces. Moreover, the platform demonstrates patterning deviation as low as $<1\%$ relative to collector geometry. In liquid management applications, we engineered spatially tailorable Janus membranes that feature dual gradients of wettability and pore structure in customized zones. These membranes exhibited specific droplet behavior and gravity-defying transport, achieving upward water migration within 372 s under capillary and hydrophilic forces. Multifunctional wearable patches further combined light responsiveness, mechanical adaptability, and site-specific liquid wicking. Incorporating 4.5 wt % MWCNT@PVA nanofibers, photothermal regions were able to hold a therapeutic temperature of 43–46.5 °C under 808 nm NIR irradiation, while maintaining spatial temperature confinement and mechanical comfort on curved skin. In addition, the printed flexible patterned circuit substrate is self-contained with electrothermal properties that are capable of maintaining a safe therapeutic heat of ~42.2 °C under body-safe voltage.

Altogether, DPMNE Tech establishes a powerful manufacturing paradigm for next-generation nanofiber-based systems, spanning reconfigurable mechanical metamaterials to programmable biomedical interfaces. Its ability to seamlessly integrate multimaterial printing, tunable mechanics, directional fluidics, and responsive functionality unlocks new opportunities in soft robotics, wearable therapeutics, and adaptive matter engineering.

EXPERIMENTAL SECTION

A Typical Fabrication Process of DPMNE Tech. Taking TPF as an example, TPU, PAN, and F127 (mass ratio of 8:1:1) were dissolved in DMF to prepare a solution with a concentration of 14 wt %. The solution was stirred for 6 h to prepare the precursor. The precursor was loaded into a syringe, maintaining a collection distance of 150 mm from the patterned PCB collector. The injection speed was set to 0.5 mL h^{-1} , with a needle voltage of 16 kV and a patterned PCB collector voltage of -1 kV . Electrospinning was performed to obtain PNFM. Detailed fabrication parameters for other precursor materials are provided in Table S1. All experiments were performed with a single electrospinning jet; multimaterial architectures were achieved by sequential precursor switching (time-multiplexed) synchronized with dynamically patterned collectors.

Fabrication and Test of Mechanical Metamaterials PNFM. Fabrication process. Using the TPF precursor solution, PNFM was collected either directly onto the patterned PCB collector module ("direct collection method") or using a $6 \mu\text{m}$ -thick PI film attached to the patterned PCB collector module ("PI film dynamic collection method"). By switching the patterned PCB collector modules beneath the PI film, additional PNFM layers were deposited in the original PNFM location, achieving the 3D stacking of nanofibers (Figure S11).

Testing process. Samples were prepared with speckle patterns using ink printing. Longitudinal tensile tests were conducted using a horizontal INSTRON 6800 testing machine, and Poisson's ratio was measured using a DIC system (CSI Vic-3D9M) (Figure S15 and Movie S2).

Fabrication and Test of Directional Liquid Transport PNFM. Fabrication process. Using a circular hole PCB collector module (Figure S19), PNFM was collected by electrospinning a 14 wt % TPF precursor for 4 min at a 150 mm collection distance, with an injection speed of 0.5 mL h^{-1} , a needle voltage of 16 kV, and a PCB collector voltage of -1 kV . Subsequently, a 12 wt % PVDF precursor was electrospun under the same conditions, but with an injection speed of 0.6 mL h^{-1} for 4 min.

Testing process. To compare the wettability of TPF and PVDF precursor materials, contact angle measurements were performed. Test samples were prepared by electrospinning each precursor for 4 min on a rectangular solid PCB collector, followed by the deposition of $1 \mu\text{L}$ water droplets. The contact angle changes over time were recorded. For the directional liquid transport performance test, $3 \mu\text{L}$ dyed water droplets were added to the TPF and PVDF sides (solid region and pore region), and the transport process was recorded using a macro lens camera (EOS 90D and EF100 mm f/2.8L Macro IS USM, Canon).

Fabrication and Test of Multifunctional Wearable Patches. Fabrication process. For the pure photothermal PNFM, 1 mL of MWCNT dispersion (WDM-M8, Beijing Boyu High-Tech Materials Co., Ltd.) was diluted with 9 mL of water, and the mixture was stirred for 1 h. Then, 5 mL of the diluted solution was mixed with PVA to prepare an MWCNT@PVA precursor with a solute concentration of 9 wt % (MWCNT concentrations in PVA were 0, 1.5, 3, 4.5, and 6 wt %). Electrospinning was performed with a collection distance of 180 mm, an injection speed of 0.6 mL h^{-1} , a needle voltage of 12 kV, a collector voltage of -1 kV , and an electrospinning time of 2 min. For the multifunctional composite PNFM, a TPF precursor was electrospun onto a peanut-shaped PCB collector for 4 min, followed by a PVDF precursor on a circular flat solid PCB collector for 2 min. Subsequently, an MWCNT@PVA precursor was electrospun onto a spiral structure PCB collector for 2 min, and finally, a PVDF precursor was electrospun onto a circular hole PCB collector for 4 min. The electrospinning parameters were consistent with those in Table S1.

UV-vis-NIR diffuse reflectance and photothermal conversion efficiency tests. Samples were prepared by electrospinning a 4.5 wt % MWCNT precursor onto a rectangular solid PCB collector for 2 min. One set was used for UV-vis-NIR diffuse reflectance tests (UV-3600, Shimadzu), and the other set was used for photothermal conversion efficiency tests. The fiber membrane samples were fixed on a sample holder, and an 808 nm laser with a power of $I = 100 \text{ mW}$ (FU808MLKH-YG2W, Shenzhen Fuzhe Technology Co., Ltd.) was

used, with an ambient temperature of 24.1°C . An infrared thermal imaging camera (A615, FLIR) was used to record the temperature changes in the laser-irradiated area. Photothermal conversion efficiency calculations refer to equation S6.

Photothermal performance tests. Samples were prepared by electrospinning MWCNT@PVA precursors with MWCNT concentrations of 0, 1.5, 3, 4.5, and 6 wt % onto a spiral-structured PCB collector for 2 min. The samples were irradiated with 808 nm near-infrared light at 400 mW. The results are shown in Figure S22A,B. Additionally, samples prepared with a 4.5 wt % MWCNT precursor were irradiated with an 808 nm laser at powers of 200, 300, 400, 500, 600, 700, and 800 mW, with the results shown in Figure S22C.

Liquid transport performance tests. For the multifunctional composite PNFM region, $3 \mu\text{L}$ water droplets were added, and the transport process was recorded using a camera and a contact angle measurement instrument (Movie S4).

Fabrication and Test of Electrically Heated PNFM. Fabrication process. The aqueous solution of silver nanowires was diluted with deionized water to 2 mg mL^{-1} . The conductive circuit with a designed bihelical pattern was then printed by using a dispenser onto the prepared polyvinyl butyral (PVB) electrospun membrane (the spinning time was 5 min, and the electrospinning parameters are shown in Table S1). The printing parameters of the dispenser are as follows: rise and fall times of 0.2 and 0.1 ms, valve opening and closing times of 1 and 15 ms, respectively, a valve opening power of 76%, a dispensing frequency of 61 Hz, nozzle distance from the membrane of 5 mm, a printing speed of 5 mm s^{-1} , and an air pressure of 0.05 MPa. Finally, another identical nanofiber membrane was attached to the patterned area to form a closed patch.

Testing process. The sample was placed on a hot plate with an initial temperature of $\sim 36.3^\circ\text{C}$ to simulate human body temperature. The probes at both ends of the sample are connected to a time relay and DC power to form a complete conductive circuit. The time relay is additionally attached to a power converter to control the on-off operation of this circuit. Temperature fluctuations in the patterned conductive circuit area of the sample are observed by varying the magnitude and switching intervals of the input voltage, and an infrared thermal imaging camera was used to focus on the test area for recording. The input voltage regularity in the heat-up test involved inputting 12, 14, 16, 18, and 20 V once each at 5 s switching intervals, while the input voltage regularity in the cycling test involved inputting 20 V ten times at 5 s switching intervals.

Nanofiber Membrane Morphology Characterization. All SEM characterizations were performed using MIRA LMS (TESCAN) with the image analysis software FIJI.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.5c12262>.

Patterned PCB-collector design, photothermal conversion efficiency calculation, electrostatic-field and NFDW time-cost simulations, electrospinning parameters, beam-width and grayscale deposition-fidelity data, nanofiber and membrane morphologies (SEM, TEM), strain-dependent Poisson's ratio data and measurement setup, light-responsive multipattern printing details, UV-vis-NIR spectra of MWCNT@PVA membranes, directional liquid transport structural details (SEM) (PDF)

Movie S1: Responsiveness of multimaterial and multipattern samples under various light spectra through DPMNE Tech (MP4)

Movie S2: Transverse strain variation of electrospun samples under longitudinal strain loading in DIC testing system (MP4)

Movie S3: Antigravity water transport in directional liquid transport PNFM (MP4)

Movie S4: Water transport and contact angle measurement in multifunctional composite PNFM (MP4)

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Notes

The authors declare no competing financial interest.

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